Hints: Polynomials – 3. Symmetric functions, Newton's identities, resultant and discriminant

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Theorem 1: order monomials lexicographically. If the leading monomial of a symmetric f is $c\,x_1^{a_1}\cdots x_n^{a_n}$, symmetry forces $a_1 \geqslant a_2 \geqslant \ldots \geqslant a_n$. Which product of the e_k has exactly this leading monomial?

Hint 2. $e_1^{a_1-a_2}e_2^{a_2-a_3}\cdots e_n^{a_n}$ has leading monomial $x_1^{a_1}\cdots x_n^{a_n}$; subtract c times it and repeat. The process stops because there are finitely many monomials of the given degree below a given one. For uniqueness, show that distinct monomials in the e_k have distinct leading monomials, so no nontrivial polynomial in the e_k vanishes.

Hint 3. Newton: for $P(t)=\prod(t-x_i)=\sum_k(-1)^k e_k t^{n-k}$ one has $\frac{tP'(t)}{P(t)}=\sum_i \frac{t}{t-x_i}=n+\sum_{k\geqslant 1}p_k t^{-k}$. Multiply by $P(t)$ and compare coefficients of t^{n-k}.

Hint 4. You should get $p_1=e_1$, $p_2=e_1^2-2e_2$, $p_3=e_1^3-3e_1e_2+3e_3$, $p_4=e_1^4-4e_1^2e_2+4e_1e_3+2e_2^2-4e_4$.

Problem 2

(a)

Hint 1. Read off $e_1=0$, $e_2=2$ from the coefficients and compute $p_2=\sum x_i^2$.

Hint 2. $p_2=e_1^2-2e_2=-4<0$; if all roots were real, p_2 would be $\geqslant 0$.

(b)

Hint 1. Use the formula for p_4 from Problem 1 with $e_1=0$, $e_2=2$, $e_3=-p$, $e_4=q$.

Hint 2. Only $2e_2^2-4e_4$ survives: $\sum x_i^4=8-4q$.

Problem 3

Hint 1. Use the partial fraction expansion $\dfrac{x^k}{P(x)}=\sum_i \dfrac{x_i^k}{P'(x_i)(x-x_i)}$, valid for $k\leqslant n-1$ when the roots are simple (reality is not needed).

Hint 2. Multiply by x and let $x\to\infty$: $\sum_i \frac{x_i^k}{P'(x_i)}=\lim_{x\to\infty}\frac{x^{k+1}}{P(x)}$.

Hint 3. The limit is 0 for $k\leqslant n-2$ and $1/a_n$ for $k=n-1$, where a_n is the leading coefficient of P.

Problem 4

Hint 1. This is the case $k=0$ of Problem 3: expand $1/p$ in partial fractions.

Hint 2. $\frac{1}{p(z)}=\sum_i \frac{1}{p'(a_i)(z-a_i)}$; multiply by z and let $z\to\infty$. The left side tends to 0 since $d\geqslant 2$.

Problem 5

Hint 1. Product formula (Res defined as the Sylvester determinant): regard both sides as polynomials in a, b and the roots α_i, β_j. If $\alpha_i = \beta_j$, the vector $(\alpha_i^{m+n-1}, \ldots, \alpha_i, 1)$ is in the kernel of the Sylvester matrix, so the determinant is divisible by every $\alpha_i - \beta_j$.

Hint 2. Compare degrees (the determinant is homogeneous of degree mn in the roots) and one coefficient, e.g. the case $\beta_1 = \cdots = \beta_m = 0$, where the determinant is easy to evaluate, to get $\operatorname{Res}(f, g) = a^m b^n \prod(\alpha_i - \beta_j) = a^m \prod_i g(\alpha_i)$. The common-root criterion follows immediately.

Hint 3. $D(ax^2 + bx + c) = b^2 - 4ac$ and $D(x^3 + px + q) = -4p^3 - 27q^2$; for the monic cubic use $D = -\operatorname{Res}(f, f') = -\prod_i f'(\alpha_i)$ with $f'(\alpha) = 3\alpha^2 + p$.

Hint 4. Eliminate y via $\operatorname{Res}_y(y^2 + x^2 - 1,\ y - x^3)$, i.e. substitute the root $y = x^3$ of the linear polynomial: $x^6 + x^2 - 1 = 0$. The solutions of the system are (x, x^3) with x a root of this.